Keerthana Devi Govindaraj

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LinkedIn | GitHub | Portfolio

Education

Northeastern University **Sep 2025 - Dec 2027**
M.S. Software Engineering Systems *Boston, MA*
SRM Institute of Science and Technology **Oct 2020 - 2024**
B.Tech, Electronics & Computer Engineering *Chennai, India*

Technical Skills

- **Languages:** Java, Python, JavaScript, TypeScript, HTML5/CSS3
- **Frontend:** React.js, Next.js, Redux, Angular, Material-UI, Responsive UI, State Management
- **Backend & APIs:** Spring Boot, FastAPI, Django, Node.js, Express.js, REST APIs, Microservices, JWT, RBAC
- **Cloud & DevOps:** AWS (EC2, RDS, S3, Lambda, DynamoDB, IAM, CloudWatch, Route53, ALB), Terraform, GitHub Actions, CI/CD, IaC, Serverless, Auto Scaling, Docker
- **Databases:** PostgreSQL, MySQL, MongoDB, SQLAlchemy, Query Optimization, Indexed Data Modeling
- **Machine Learning & AI:** TensorFlow, PyTorch, OpenCV, NLP, LLM Integration, Gemini API
- **Practices:** Git, Agile, SDLC, OOP, Performance Optimization, API Testing

Experience

Linedata - Product Analyst **Sep 2026 - Present**
Boston, MA

- Contributing to the modernization of Linedata's Web Admin module, collaborating with product and engineering teams to improve the platform's architecture, usability, and maintainability.
- Supporting product development and Agile initiatives across asset-management software, working with product managers, architecture, R&D, and Scrum teams to drive feature and platform improvements.

FITA Academy - Software Engineer **Jan 2025 - Aug 2025**
Chennai, India

- Led end-to-end development and deployment of a full-stack Student Management System serving 500+ students, automating enrollment, attendance, academic reporting, and progress tracking, replacing spreadsheet-based operations.
- Designed and optimized scalable Java Spring Boot microservices and React UIs, building 20+ secure REST APIs with JWT authentication, RBAC, MongoDB, and CI/CD pipelines, achieving sub-100ms p95 response times and improving release velocity by 30%.

Aican Automate - Machine Learning Engineer **Jan 2023 - Mar 2023**
Chennai, India

- Built and evaluated NLP-based support-ticket classification systems using TF-IDF, N-gram feature engineering, Logistic Regression, and Random Forest, achieving 87% precision and significantly reducing manual triage effort.
- Productionized ML workflows with Docker-based containerization, creating reproducible training and deployment pipelines that enabled 5+ engineers to retrain, validate, and deploy models with minimal overhead.

Projects

SchemaLens - Legacy PostgreSQL Documentation and Lineage | Python, TypeScript, PostgreSQL, PL/pgSQL, MCPLive | Code

- Built a tool that makes legacy PostgreSQL databases understandable: searchable docs, read/write lineage parsed directly from PL/pgSQL, and LLM explanations cross-checked against static analysis, with accuracy scored against a hand-labeled benchmark in CI.
- Shipped an MCP server so AI coding assistants answer schema questions from the real scan instead of guessing, with a privacy model that queries only system catalogs and never reads row data.

HeatSink - Seawater-Cooled Data Center Siting Tool | TypeScript, React, Vite, Vitest Live | Code

- Built a full-stack screening tool that maps coastal temperatures and models thermal-plume impact to identify where a seawater-cooled AI data center does the least harm to the ocean, with plain-English risk verdicts for planners.
- Engineered interactive geospatial visualizations and configurable modeling sliders in TypeScript, backed by unit and property-based tests and a CI pipeline for reliable, reproducible builds.

Cloud-Native Serverless Web Application | Python, FastAPI, PostgreSQL, Terraform, AWS Infra | App

- Automated provisioning of production-style AWS infrastructure using 3,000+ lines of Terraform across VPCs, EC2 Auto Scaling Groups, RDS Multi-AZ, ALB, and Route53, enabling repeatable deployments via GitHub Actions CI/CD.
- Designed a serverless verification workflow using SNS, Lambda, and DynamoDB with token expiration and deduplication logic, and hardened security with IAM, KMS encryption, Secrets Manager, BCrypt, and ACM-managed TLS.

Eye-Tracking Based ASD Diagnosis (Published) | Python, TensorFlow, CNN, PCA, OpenCV Code

- Applied transfer learning and PCA-based dimensionality reduction on CNN models to classify ASD risk across 10,000+ eye-tracking samples, improving prediction accuracy by 20% over baseline models.
- Built an OpenCV preprocessing pipeline that converted 10,000+ variable-frame-rate recordings into model-ready tensors, published as a research paper.